

Injury Risk in University Soccer Players

Predicting Injury Risk with Data

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Data Background and Overview

- **Dataset contains data for 800 Chinese university soccer players**
- **Includes 18 variables covering**
 - Basic information (age, height, weight, etc.)
 - Physical fitness (agility, sprint speed, balance, flexibility, etc.)
 - Training factors (training hours, matches played, etc.)
 - Lifestyle Habits (sleep, stress, nutrition, warmup routine)
 - Injury outcome
- **Data obtained from Kaggle, originates from a comprehensive study done by the university using medical records, coach reports, and student surveys. Published in Scientific Reports (2025)**

Primary Goal

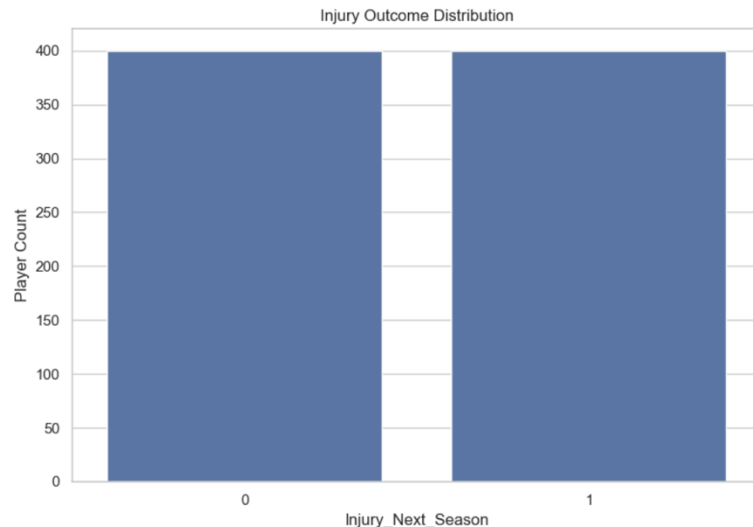
- The primary goal of this project is to explore how training load, previous injury history, physical fitness metrics, and lifestyle habits relate to injury risk in university soccer players.
- This project aims to identify which factors appear most strongly associated with injuries and how those insights could support evidence-based prevention strategies.
- By understanding these relationships, sports programs can develop targeted interventions to reduce injury rates and protect athlete health.

Exploratory Data Analysis

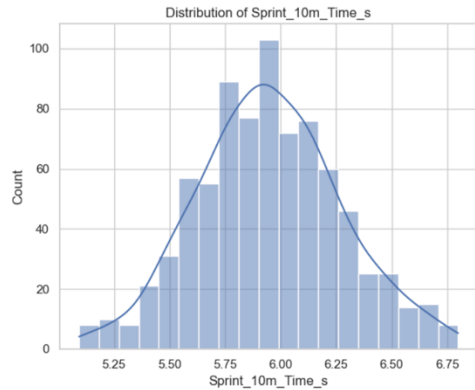
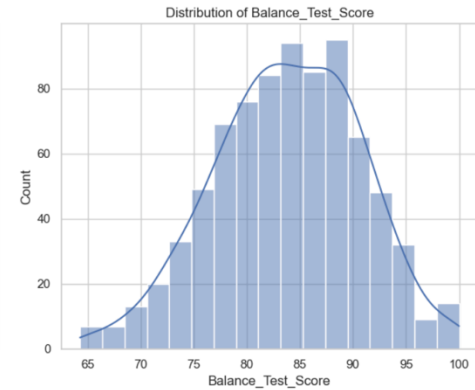
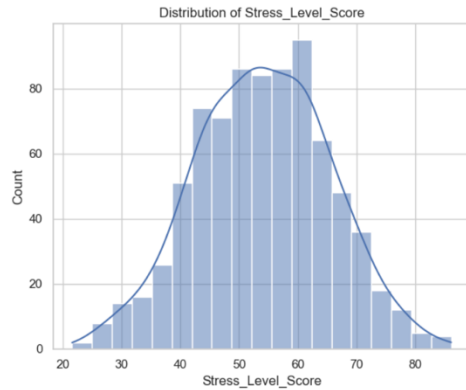
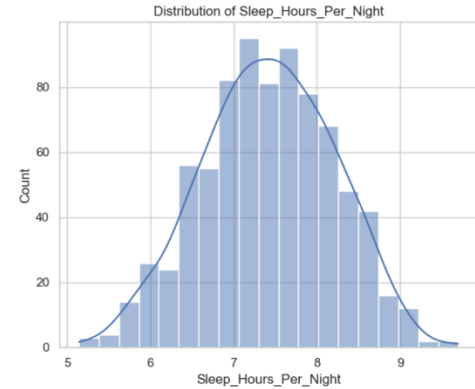
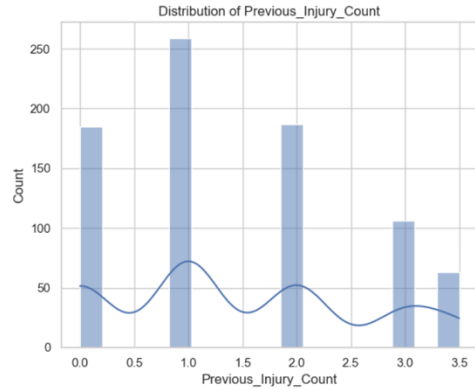
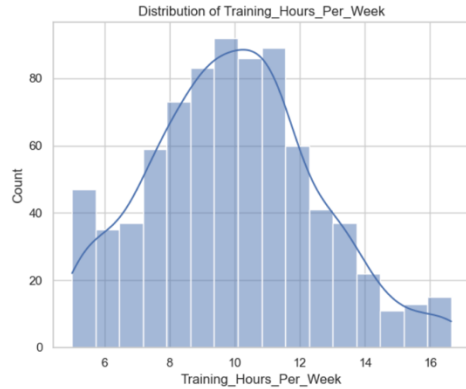
	count	mean	std	min	25%	50%	75%	max
Age	800.00	21.14	1.99	18.00	19.00	21.00	23.00	24.00
Height_cm	800.00	177.41	7.05	159.50	173.00	177.00	182.00	195.50
Weight_kg	800.00	73.23	9.91	45.00	66.00	73.00	80.00	101.00
BMI	800.00	23.37	3.66	14.35	20.79	23.13	26.02	33.88
Training_Hours_Per_Week	800.00	9.94	2.58	5.00	8.13	9.90	11.54	16.65
Matches_Played_Past_Season	800.00	22.33	10.31	5.00	13.00	22.00	32.00	39.00
Previous_Injury_Count	800.00	1.46	1.13	0.00	1.00	1.00	2.00	3.50
Knee_Strength_Score	800.00	74.94	6.65	56.63	70.43	75.00	79.63	93.43
Hamstring_Flexibility	800.00	79.15	6.76	60.52	74.50	79.19	83.81	97.79
Reaction_Time_ms	800.00	249.44	22.48	187.57	234.09	249.13	265.11	306.73
Balance_Test_Score	800.00	83.84	6.90	64.30	79.04	84.16	88.88	100.00
Sprint_10m_Time_s	800.00	5.95	0.33	5.09	5.73	5.94	6.16	6.80
Agility_Score	800.00	78.36	8.71	55.55	72.68	78.34	84.09	100.00
Sleep_Hours_Per_Night	800.00	7.42	0.79	5.14	6.85	7.42	7.99	9.70
Stress_Level_Score	800.00	54.04	11.42	21.57	45.78	54.05	61.91	86.11
Nutrition_Quality_Score	800.00	74.38	9.32	50.00	67.81	74.36	80.50	99.54
Warmup_Routine_Adherence	800.00	0.60	0.49	0.00	0.00	1.00	1.00	1.00

Descriptive Statistics for Numerical Variables

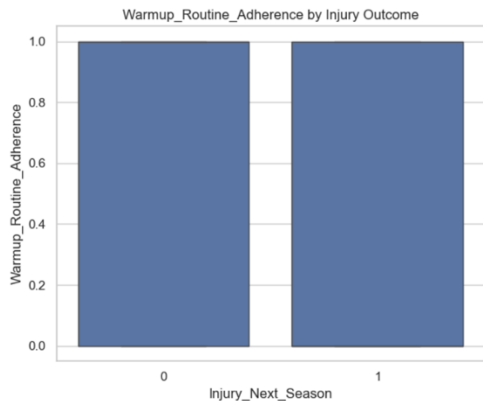
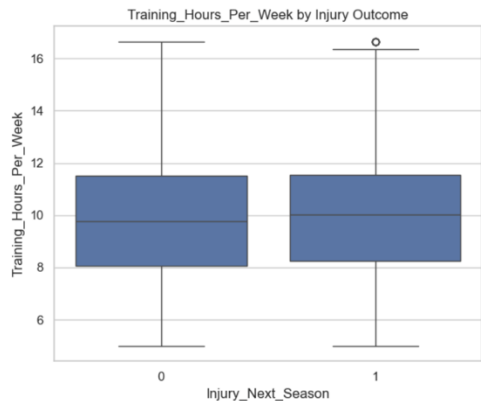
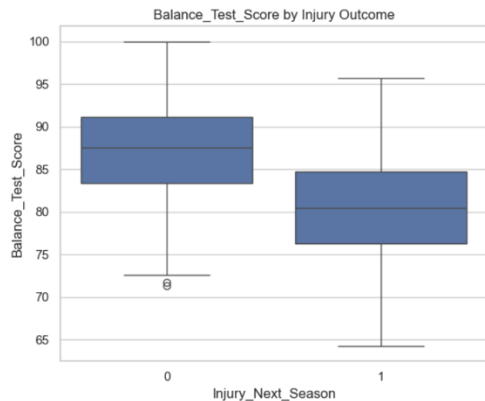
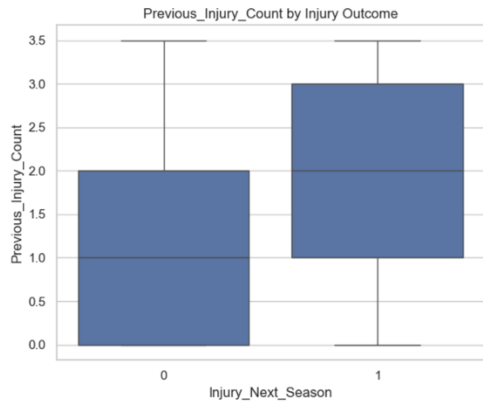
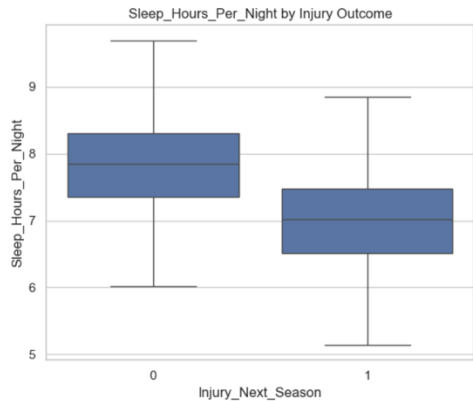
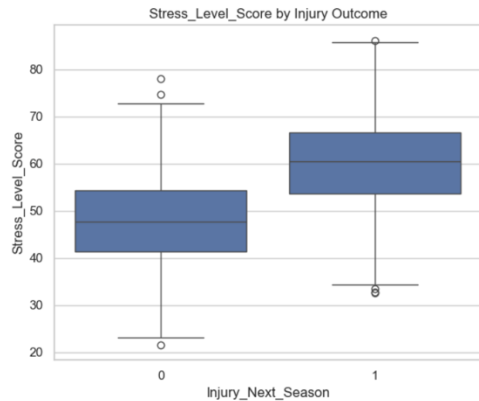
Each variable has a numeric value associated with it to compare each player effectively



Key Variable Value Distribution

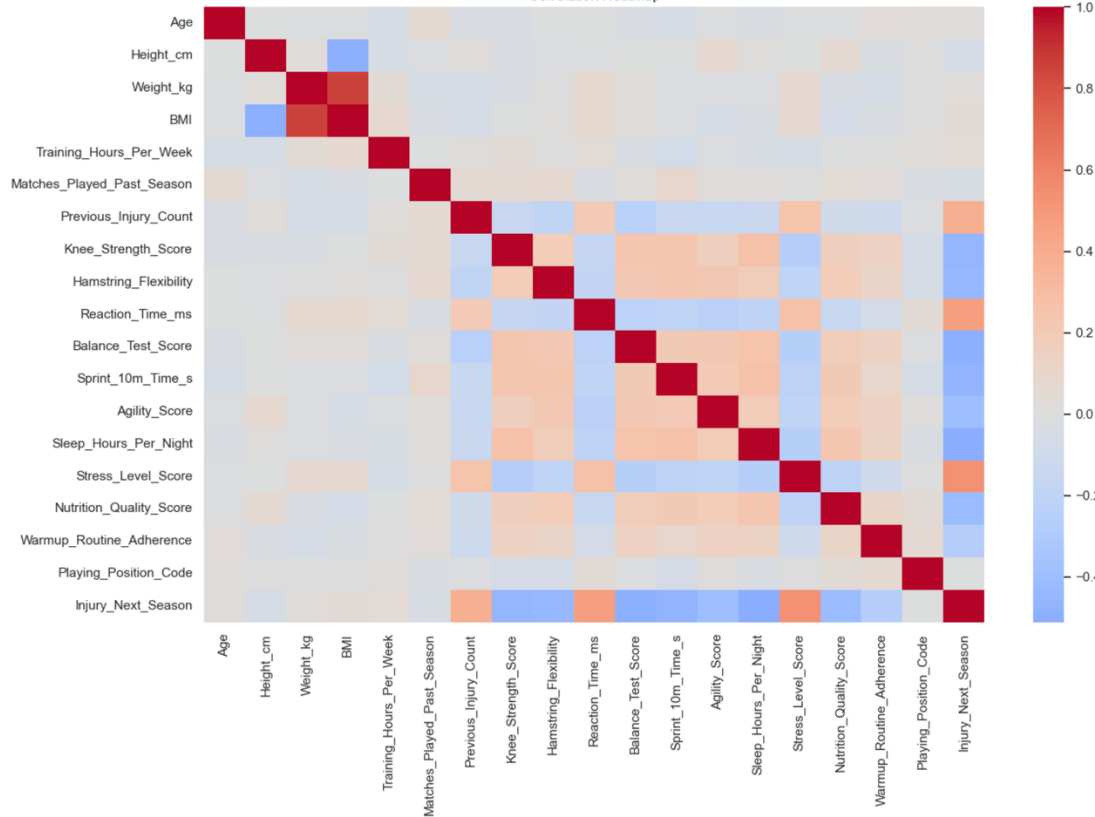


Boxplots



Correlations

Correlation Heatmap

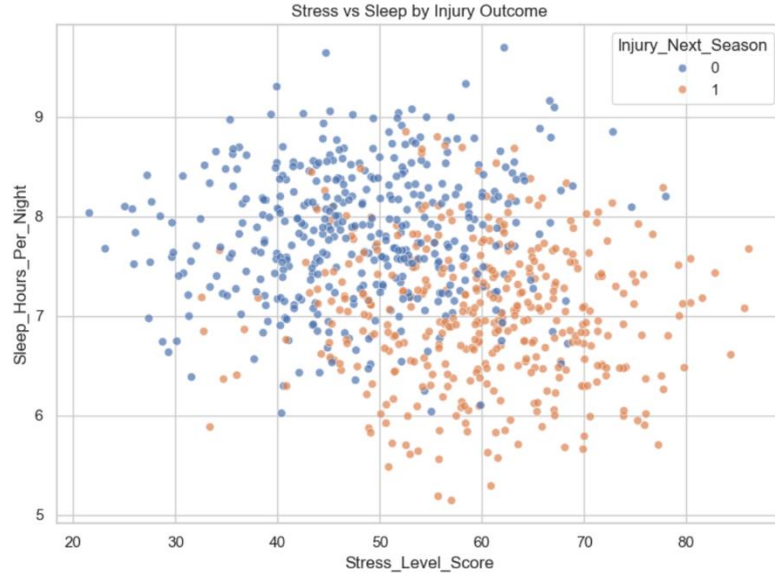


Injury_Next_Season	0	1
Training_Hours_Per_Week	9.86	10.02
Previous_Injury_Count	1.03	1.90
Sleep_Hours_Per_Night	7.82	7.01
Stress_Level_Score	47.99	60.09
Balance_Test_Score	87.26	80.43
Warmup_Routine_Adherence	0.73	0.47

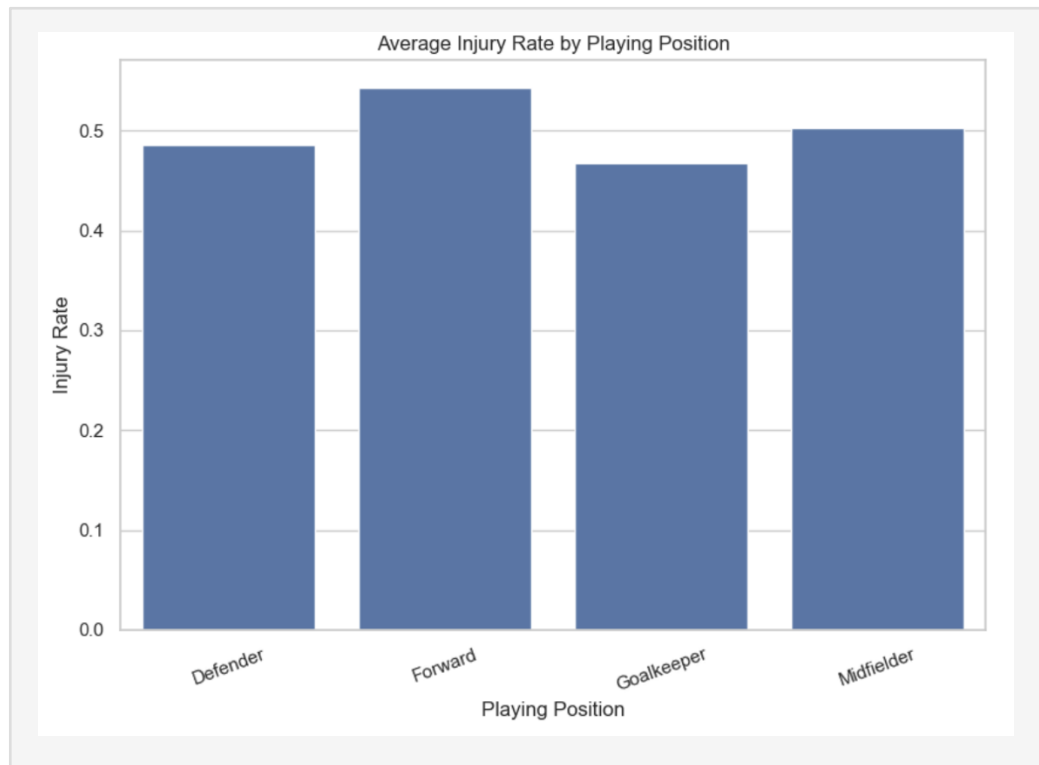
Red Boxes = Positive Correlation
Ex. High stress levels correlate with injury

Blue Boxes = Negative Correlation
Ex. Poor nutrition leads to injury

Scatterplots



Positional Risk



Factors that lead to injury...

Lifestyle: High stress levels, inadequate sleep, and poor nutrition

History: Previous injury count strongly predictive

Physical: Low lower body balance, flexibility, and strength

Behavioral: Inconsistent training and poor warmup adherence

Positional: Forwards and Midfielders at a higher risk of injury than Defenders and Goalkeepers

- **Privacy and Data Security**
 - Athlete health is highly sensitive
 - High security measures required
- **Fairness and Equity**
 - Avoid stigmatization of “high-risk” players
- **Responsible Implementation**
 - Avoid overreliance on prediction models
 - Informed consent for data use
 - Transparent communication with athletes

- **This research is of high value to athletes and sport clubs to see what they should and shouldn't do in order to prevent injury**
- **Business is better for sport clubs when their athletes are healthy and playing the sport for fans to watch. Teams will bring more fans in when their favorite athletes are healthy**
- **In the future...**
 - Potentially stretch research to multiple seasons
 - Diversify populations
 - Implementation of research